01 - Business Problem and Objective

Customer Churn Prediction | Repository-based interview study guide

Source: https://github.com/YuvarajGarla/Customer-churn-prediction

1. Project context

Customer churn is the situation in which an existing customer stops using a company's products or services. For a telecom provider, churn may mean disconnecting a mobile connection, broadband subscription, or another paid service.

The business problem is to identify customers who are at elevated risk of leaving, early enough for the company to consider an appropriate retention action. Acquiring a replacement customer can require marketing, onboarding, and sales expenditure; retaining a suitable existing customer may therefore be economically attractive.

2. Machine-learning objective

This is a supervised binary-classification problem. The target is Churn: Yes is encoded as 1 and No as 0. The model attempts to estimate P(Churn = 1 | customer information), using demographic, account, service, tenure, and billing attributes.

$$Y = 1 \text{ if the customer churns; } Y = 0 \text{ otherwise}$$

$$\text{Model objective: estimate } P(Y = 1 \mid X), \text{ where } X \text{ is the vector of customer features.}$$

3. Business use

- Prioritize customers with high predicted churn risk for review or outreach.
- Tailor retention offers rather than applying the same intervention to every customer.
- Investigate service or contract characteristics associated with churn.
- Measure whether retention interventions work through a properly designed experiment.

Important: A prediction is not a causal explanation. The model estimates risk from historical associations; it does not prove why a customer leaves or guarantee that an offer will prevent churn.

4. Interview-ready explanation

The objective of this project was to predict customer churn for a telecom company. Customer churn refers to customers discontinuing their telecom services. Since retaining existing customers is an important business priority, we wanted to identify customers who were likely to leave using their demographic information, subscription details, service usage, tenure, and billing information. The predictions could help the business prioritize high-risk customers for proactive retention measures.

5. Questions to prepare

Q. Why is this a classification problem?

The outcome is a discrete label with two classes: churn and non-churn. The model can estimate a probability and then assign a class using a decision threshold.

Q. What is the business value of churn prediction?

It helps prioritize limited retention resources toward customers who may be at risk. The value depends on whether interventions change behaviour enough to justify their cost.

Q. Does a high churn probability tell us why a customer leaves?

No. It is a predictive risk score, not a causal explanation. Explanation and intervention-effect estimation need additional analysis.